

03/25 /2026

Quasicoherent sheaves

on affine toric stacks I

Last time Proof overview, the combinatorial model, background on idempotent algebras (for a little later).

Today Start the proof that $\mathrm{QCoh}(X_\sigma/T) \simeq \mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp})$

Recall For a lattice N and cone $\sigma \in N_{\mathbb{R}}$:

(1) $\mathrm{Poly}(M)$ is the poset of Minkowski sums

(polytope) + (cone)

in $M_{\mathbb{R}}$. This is symmetric monoidal under Minkowski sum
- Cones are idempotent algebras

(2) $\Theta(\sigma) := \{m + \sigma^\vee \mid m \in M\} \subset \mathrm{Mod}_{\sigma^\vee}(\mathrm{Poly}(M))$

This is sym. mon. under Minkowski sum with unit $\sigma^\vee$

(3) For an inclusion $\iota: \sigma \hookrightarrow \tau$ of cones, we have a sym mon functor

$$\begin{array}{ccc} \Theta(\iota): \Theta(\tau) & \longrightarrow & \Theta(\sigma) \\ m + \tau^\vee & \longmapsto & m + \sigma^\vee \end{array} \quad \left. \vphantom{\begin{array}{ccc} \Theta(\iota): \Theta(\tau) & \longrightarrow & \Theta(\sigma) \\ m + \tau^\vee & \longmapsto & m + \sigma^\vee \end{array}} \right\} \text{ just } + \sigma^\vee$$

Digression: Relative tensor products

> We should have said this earlier, but forgot to.

Recall. Let A be an ordinary commutative ring and $M, N \in \text{Mod}_A^{\text{fd}}$. Then the tensor product $M \otimes_A N$ can be extracted from only knowing $\otimes_{\mathbb{Z}}$ as the coequalizer of

$$M \otimes_{\mathbb{Z}} A \otimes_{\mathbb{Z}} N \begin{array}{c} \xrightarrow{(\text{mod str for } M) \otimes \text{id}} \\ \xrightarrow{\text{id} \otimes (\text{mod str for } N)} \end{array} M \otimes_{\mathbb{Z}} N$$

> Let's generalize this to the higher-categorical setting. As usual, we'll need to replace coequalizers by colimits of simplicial objects.

Def. A colimit of a simplicial diagram X in an ∞ -category is called a geometric realization, and often denoted by $|X|$.

The bar construction. Let $\mathcal{C}$ be a symmetric monoidal ∞ -category, $A \in \text{Catg}(\mathcal{C})$, and $M, N \in \text{Mod}_A(\mathcal{C})$. There is a natural simplicial object $\text{Bar}_A(M, N)$ informally defined as follows

$$\begin{array}{ccccc} & \xrightarrow{(\text{mod str for } M) \otimes \text{id}} & & \xrightarrow{(\text{mod str for } M) \otimes \text{id}} & \\ & \leftarrow \text{id} \otimes \text{unit} \otimes \text{id} \otimes \text{id} \leftarrow & & \leftarrow \text{id} \otimes \text{unit} \otimes \text{id} \leftarrow & \\ \cdots M \otimes A \otimes A \otimes N & \xrightarrow{\text{id} \otimes \text{mult} \otimes \text{id}} & M \otimes A \otimes N & \xrightarrow{\text{id} \otimes \text{unit} \otimes \text{id}} & M \otimes N \\ & \leftarrow \text{id} \otimes \text{id} \otimes \text{unit} \otimes \text{id} \leftarrow & & \xrightarrow{\text{id} \otimes (\text{mod str for } N)} & \\ & \xrightarrow{\text{id} \otimes (\text{mod str for } N)} & & & \end{array}$$

Thm. Let $\mathcal{C}$ be a symmetric monoidal ∞ -category that admits geometric realizations and assume that $\otimes_{\mathcal{C}} : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ preserves geometric realizations separately in each variable. Let $A \in \text{Catg}(\mathcal{C})$. Then $\text{Mod}_A(\mathcal{C})$ admits a symmetric monoidal Str with unit A whose underlying object is

$$M \otimes_A N := |\text{Bar}_A(M, N)|.$$

Moreover, $\otimes_A$ preserves geom. realizations separately in each var.

The unstable analogue of $\mathrm{QCoh}(X_\sigma/T)$

Recall. Let N be a lattice, $\sigma \subset N_{\mathbb{R}}$ a cone, and $S_\sigma = M \cap \sigma^\vee$.
Then $T = \mathrm{Spec}(S\langle M \rangle)$ and $X_\sigma = \mathrm{Spec}(S\langle S_\sigma \rangle)$. By construction

$$\mathrm{QCoh}(X_\sigma/T) \cong \lim \left(\begin{array}{c} \mathrm{QCoh}(X_\sigma) \rightrightarrows \mathrm{QCoh}(X_\sigma \times T) \rightrightarrows \cdots \\ \text{SI} \qquad \qquad \text{SI} \\ \mathrm{Mod}_{S\langle S_\sigma \rangle}^{(Sp)} \qquad \mathrm{Mod}_{S\langle S_\sigma \times M \rangle} \end{array} \right).$$

That is, $\mathrm{QCoh}(X_\sigma/T)$ is the limit of the diagram we get by applying QCoh with pullback functoriality to the diagram

$$\cdots \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} X_\sigma \times T \times T \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} X_\sigma \times T \longrightarrow X_\sigma$$

defining the action.

But recall that this diagram is obtained by applying $\mathrm{Spec}(S\langle - \rangle)$ to the diagram in $\mathrm{CMon}(\mathrm{Set})$ defining the coaction of M on S_σ .

$$\cdots \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} S_\sigma \times M \times M \quad \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} S_\sigma \times M \quad \longleftarrow S_\sigma$$

Construction. Passing to module ∞ -categories with functoriality extension of scalars, we obtain a cosimplicial diagram

$$\cdots \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} \text{Mod}_{S_\sigma \times M \times M}(A_n) \quad \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} \text{Mod}_{S_\sigma \times M}(A_n) \quad \longleftarrow \text{Mod}_{S_\sigma}(A_n)$$

> we write $\text{Mod}_{S_\sigma}(A_n)^M$ for the limit of the above diagram

> we write $\text{Mod}_{S_\sigma}(\text{Set})^M$ for the limit of the analogous diagram with A_n replaced by Set .

- Recall that we can replace Δ by $\Delta_{\leq 2}$, and explicitly describe the limit (which involves the cocycle condition)

Construction (the tautological) object of $\text{Mod}_{S_\sigma}(\text{Set})^M$. Let's refine the S_σ -module M (via the inclusion) we need to provide an isomorphism

$$f: M \otimes_{S_\sigma} (S_\sigma \times M) \xrightarrow{\sim} M \otimes_{S_\sigma} (S_\sigma \times M)$$

$S_\sigma \rightarrow S_\sigma \times M$ is (id, inclusion) relative tensor $S_\sigma \rightarrow S_\sigma \times M$ is (id, 0)

Satisfying the cocycle condition. We take the isomorphism to be

$$(m, s, n) \longmapsto (m, s, m+n)$$

Exercise. Unpack this definition and check that f satisfies the cocycle condition

Idea. This object is an unstable refinement of the $*$ -pushforward of the structure sheaf $\mathcal{O}_{T/T}$ along the open immersion

$$T/T \hookrightarrow X_\sigma/T$$

> This is literally true, but to make sense of it, we need a comparison of $\text{Mod}_{S_r}(A_n)^M$ to $\text{QCoh}(X_\sigma/T)$

> We'll need a technical digression

Prop Let $\mathcal{C}$ and $\mathcal{D}$ be symmetric monoidal ∞ -categories admitting geometric realizations and let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a sym mon functor

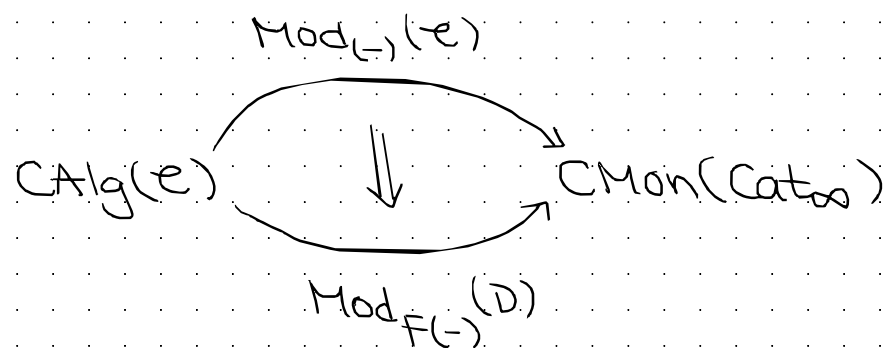
Assume:

- (1) The tensor products in $\mathcal{C}$ and $\mathcal{D}$ preserve geometric realizations separately in each variable
- (2) F preserves geometric realizations

Then for each $A \in \text{Catg}(\mathcal{C})$, the induced functor

$$F: \text{Mod}_A(\mathcal{C}) \longrightarrow \text{Mod}_{F(A)}(\mathcal{D})$$

is naturally symmetric monoidal. Moreover, termwise application of F defines a natural transformation



Ex. The free module functor $S \langle - \rangle : \text{An} \rightarrow \text{Sp}$ is symmetric monoidal and preserves all colimits.

> So for $A \in \text{CMon}(\text{An})$, we get a symmetric monoidal functor

$$S \langle - \rangle : \text{Mod}_A(\text{An}) \longrightarrow \text{Mod}_{S \langle A \rangle}(\text{Sp})$$

> Applying the functoriality of the above result to the diagram

$$\dots \quad \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} S_\sigma \times M \times M \quad \begin{array}{c} \longleftarrow \\ \longleftarrow \\ \longleftarrow \\ \longleftarrow \end{array} S_\sigma \times M \quad \begin{array}{c} \longleftarrow \\ \longleftarrow \end{array} S_\sigma$$

We get a natural transformation of diagrams of sym. mon. ∞ -cats (indexed by $\mathbb{A}$)

$$\begin{array}{ccc} \mathrm{Mod}_{S_\sigma \times M^{(\bullet-1)}}(A_n) & \longrightarrow & \mathrm{Mod}_{S\langle S_\sigma \times M^{(\bullet-1)} \rangle}(Sp) \\ & & \downarrow S \\ & & \mathrm{QCoh}(X_\sigma \times T^{(\bullet-1)}) \end{array}$$

> Taking limits, we obtain a functor

$$\Theta[-]: \mathrm{Mod}_{S_\sigma}(A_n)^M \longrightarrow \mathrm{QCoh}(X_\sigma/T)$$

Now we want to write down a sym. mon. functor

$$\mathrm{Fun}(\Theta(\sigma), Sp) \longrightarrow \mathrm{Mod}_{S_\sigma}(A_n)^M$$

> The composite will be our desired functor.

The universal property of Day convolution

> In fact, it will suffice to construct a functor out of $\mathcal{O}(\mathcal{C})$

Ntn. For a small ∞ -category $\mathcal{C}$, write h for the composite

$$\mathcal{C} \xrightarrow{\mathcal{L}} \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{A}_n) \xrightarrow{\$ \dashv \circ -} \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}_p).$$

Thm. Let $\mathcal{C}$ be a small sym. mon ∞ -cat and let $\mathcal{D}$ be a sym. mon ∞ -cat with colimits such that the tensor product preserves colimits separately in each variable. Then:

(1) Restriction along $\mathcal{L}$ defines an equivalence

$$\mathrm{Fun}^{\otimes, \mathrm{colim}}((\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{A}_n), \otimes_{\mathrm{Day}}), \mathcal{D}) \xrightarrow{\sim} \mathrm{Fun}^{\otimes}(\mathcal{C}, \mathcal{D})$$

(2) If $\mathcal{D}$ is stable, restriction along h defines an equiv

$$\mathrm{Fun}^{\otimes, \mathrm{colim}}((\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}_p), \otimes_{\mathrm{Day}}), \mathcal{D}) \xrightarrow{\sim} \mathrm{Fun}^{\otimes}(\mathcal{C}, \mathcal{D})$$

Combinatorial to unstable

Construction. Let N be a lattice and $\sigma \in N_{\mathbb{R}}$ a cone. Define a functor

$$\phi_{\sigma}: \Theta(\sigma) \longrightarrow \text{Mod}_{S_{\sigma}}(\text{Set})^M$$

as follows. For $V \in \Theta(\sigma)$ (which is an integral translate of $\sigma^{\vee}$), note that $V \cap M$ is a subobject of M in $\text{Mod}_{S_{\sigma}}(\text{Set})$ and Set

$$\phi_{\sigma}(V) := (V \cap M, f|_{(V \cap M) \otimes_{S_{\sigma}} (S_{\sigma} \times M)}),$$

where $f: M \otimes_{S_{\sigma}} (S_{\sigma} \times M) \xrightarrow{\sim} M \otimes_{S_{\sigma}} (S_{\sigma} \times M)$ is the iso defining the tautological object of $\text{Mod}_{S_{\sigma}}(\text{Set})^M$ from before.

- > By the def of f , it is clear that f restricts to $(V \cap M) \otimes_{S_{\sigma}} (S_{\sigma} \times M)$.
- > For an inclusion $i: V \hookrightarrow W$, we define $\phi_{\sigma}(i)$ to be the induced inclusion $V \cap M \hookrightarrow W \cap M$.

- Note that since the structure isomorphisms for $\phi_\sigma(V)$ and $\phi_\sigma(W)$ are both inherited from the tautological object (M, f) , this actually defines a map $\phi_\sigma(V) \rightarrow \phi_\sigma(W)$.

Exercise Check that $\phi_\sigma: \Theta(\sigma) \rightarrow \text{Mod}_S(\text{Set})^M$ naturally refines to a Sym mon functor with respect to Minkowski Sum.

> Note that $\phi_\sigma(\sigma^\vee) = \sigma^\vee \cap M = S_\sigma$ is the unit.

Upshot. We have functors

$$\Theta(\sigma) \xrightarrow{\phi_\sigma} \text{Mod}_S(\text{Set})^M \hookrightarrow \text{Mod}_S(\text{An})^M \xrightarrow{\Theta[-]} \text{QCoh}(X_\sigma/T)$$

We know ϕ_σ and $\Theta[-]$ are Sym monoidal.

 The inclusion $\text{Mod}_S(\text{Set})^M \hookrightarrow \text{Mod}_S(\text{An})^M$ is **not** symmetric monoidal! The problem is that $\text{Mod}_S(\text{Set}) \hookrightarrow \text{Mod}_S(\text{An})$ isn't Sym. mon.

Question. So it seems we have a problem. How do we resolve this?

Observation. Let $S \in \mathbf{CMon}(\mathbf{Set})$. Let $\text{Mod}_S^{\text{free}}(\mathbf{An}) \subset \text{Mod}_S(\mathbf{An})$ denote the full subcategory spanned by coproducts of S .

- > Then $\text{Mod}_S^{\text{free}}(\mathbf{An}) \subset \text{Mod}_S(\mathbf{An})$ is closed under the relative tensor product $\otimes_S$.
- > Moreover, $\text{Mod}_S^{\text{free}}(\mathbf{An})$ coincides with $\text{Mod}_S^{\text{free}}(\mathbf{Set}) \subset \text{Mod}_S(\mathbf{Set})$.

Observation. For a lattice N and cone $\sigma \in N_{\mathbb{R}}$, write

$$\text{Mod}_{S_\sigma}^{\text{free}}(\mathbf{Set})^M = \lim_{[n] \in \Delta} \text{Mod}_{S_\sigma \times M^{\times(n-1)}}^{\text{free}}(\mathbf{Set})$$

That is, $\text{Mod}_{S_\sigma}^{\text{free}}(\mathbf{Set})^M \subset \text{Mod}_{S_\sigma}(\mathbf{An})^M$ is the full subcategory spanned by those objects such that each term is free.

- > Then the inclusion $\text{Mod}_{S_\sigma}^{\text{free}}(\mathbf{Set})^M \hookrightarrow \text{Mod}_{S_\sigma}(\mathbf{An})^M$ is symmetric monoidal.

Observation. For each $V \in \Theta(\sigma)$, we have $V = m + \sigma^V$ for some $m \in M$. So

$$V \cap M = m + (\sigma^V \cap M) = m + S_\sigma.$$

As an S_σ -module, this is isomorphic to S_σ , hence free.

Upshot. The functor

$$\Theta(\sigma) \xrightarrow{\phi_\sigma} \text{Mod}_{S_\sigma}^{\text{free}}(\text{Set})^M \hookrightarrow \text{Mod}_{S_\sigma}(\text{An})^M \xrightarrow{\mathcal{G}[-]} \text{QCoh}(X_\sigma/T)$$

is sym mon. Hence this induces a sym. mon. left adjoint

$$(\text{Fun}(\Theta(\sigma)^{\text{op}}, \text{Sp}), \otimes_{\text{Day}}) \longrightarrow \text{QCoh}(X_\sigma/T).$$